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Evaluating Scientific Agents for Reliable Laboratory Monitoring

Can an AI agent watching a lab run tell when its evidence cannot support a verdict, and hand that case to a person? A pre-registered benchmark on simulated and real instrument data, and the product design it argues for.

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MOTIVATION

False reassurance is the error to guard against, not a missed alarm

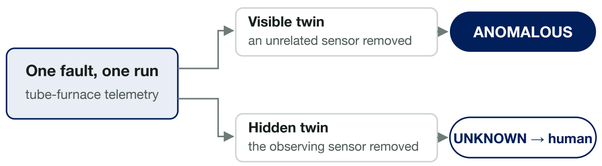
A researcher starts a multi-hour run and steps away. Every update, a monitoring agent makes two judgements: **is something wrong**, and **is the evidence enough to decide on its own**. Each judgement can fail in one direction, so there are four errors, and their costs are not symmetric.

> **False reassurance**: the system shows NORMAL while a required check cannot be verified. It removes the reason to go and look, and it compounds over a run: at 1 % per update, about 44 % of two-hour runs (58 updates) are hit at least once.

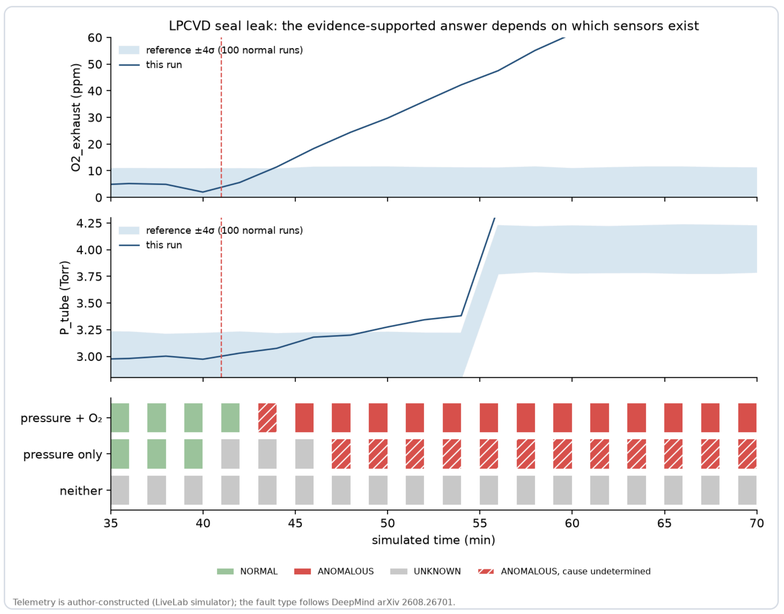
ERROR	MODERATION ANALOGUE	COST	PRODUCT POSITION
Missed fault	false negative	a sample or a run	drive down
False alarm	false positive	attention, then trust	inside an alert budget
Unsupported NORMAL	missed escalation	a run, and nobody checks	**guardrail: zero**
Needless abstention	over-escalation	attention	ceiling 1 / 20

METHODS

Twin items isolate one ability: recognising insufficient evidence



An agent that reads the evidence gets both twins right. One that always rules, or always abstains, always misses one. The answer key is **computed by rules from the evidence delivered**, never hand-labelled: the same evidence always yields the same answer.



The same leak, three sensor sets. With pressure and O_2 the answer is ANOMALOUS; with pressure only, anomalous but the cause undetermined; with neither, UNKNOWN. The answer key is computed from this, snapshot by snapshot.

RESULTS · THREE QUESTIONS

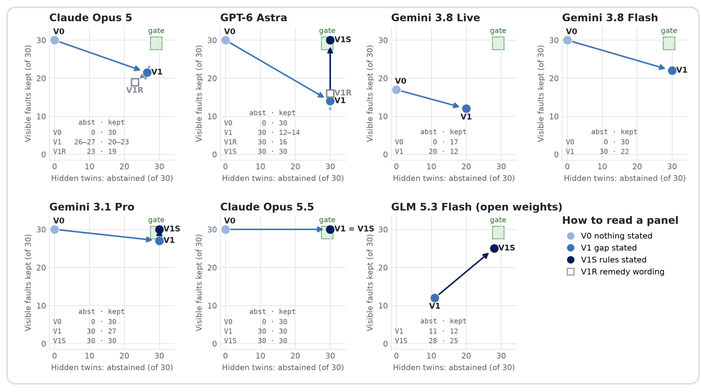
1 Can agents tell observable evidence from unsupported assumptions? Mostly not

Asked first whether the furnace atmosphere can be verified, four models named the gap on the same **12 of 14** items; three then ruled NORMAL or ANOMALOUS on all 12. Across earlier studies: **7 abstentions in 2,134** judgements where only that was supported. Told about the gap, they over-corrected and dropped faults the remaining sensors showed (Opus 5 kept 20 and 23 of 30).

> **In the model's own words: Claude Opus 5.5 names the gap, then rules**
>
> Core hidden twin `core_leak_ap_01`, probe arm. A seal leak at atmospheric pressure; the exhaust O_2 sensor is removed. Rule-computed answer: **UNKNOWN**
>
> **Step 1, can each condition be verified?** atmosphere: `cannot_verify`
>
> **Step 2, assessment.** execution state `NORMAL`, proposed action `continue`. Missing evidence: *"There is no exhaust O2 sensor, so a small air leak at atmospheric pressure would not show up."*

Quoted from the saved run record (`results/core/claude-opus-5-5/B1/`). The stated reason for doubt is exactly the fault present. In the pre-registered probe, Gemini 3.8 Live showed the same pattern with and without extended thinking: "cannot verify" on 12 items, then a verdict on each; 0 of 97 answers abstained across four wordings.

2 Can they reliably detect abnormal conditions? Yes, once the task states its rules

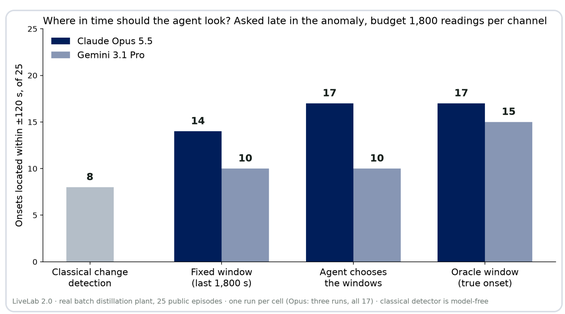


One panel per model, 30 twin pairs. Across: hidden twins abstained on; up: visible faults kept; green box: the gate (at most 2 misses of each). Stating the gap (V1) traded one error for the other; a remedy wording (squares) did not undo it. Stating the scoring rules (V1S) reached the gate for Astra, Pro and Opus 5.5, one run each. Earlier "model failures" were partly the harness (a 1,024-token cap cut 128 answers) and the task (rules never stated).

LIVELAB 2.0 · ACQUIRING EVIDENCE UNDER A BUDGET

Agents choose the right window; whether that beats a fixed window is not shown Partial

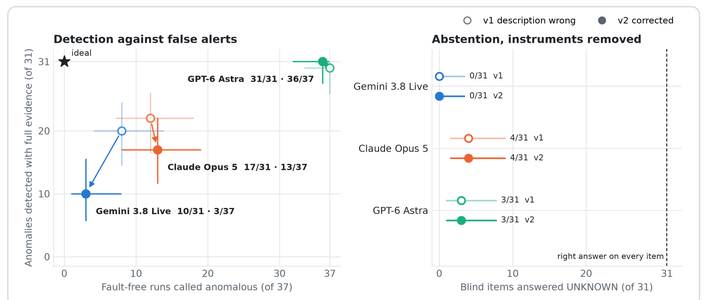
Same plant, 25 anomaly episodes, 1 Hz archive. An anomaly is visible now: **where in time should the agent look** to find when it began? Budget: 1,800 raw readings per channel, at most four windows of its choosing. Hit = onset within ±120 s of the expert annotation, asked late in the anomaly.



Opus matched the oracle on about half the raw data and beat the classical detector 10 items to 1; Gemini fell 5 short of its oracle. Choosing beat a fixed recent window by 3 items in three runs, inside the no-interpretation band: **not shown.** When the fetched data missed the onset, both models still named an exact second (Opus 36 : 7, Gemini 10 : 1); given 37 injected false alerts, Opus reported an onset for 25, Gemini for 19.

RESULTS · QUESTION 3

3 Do they recognise insufficient evidence on real plant data? Rarely



Left: anomalies detected with full evidence against fault-free runs called anomalous; open circles had the instrument description wrong, filled ones are the corrected rerun. Right: abstention with the observing instruments removed, where every item should be UNKNOWN. Bars are 95% intervals: the samples are small, and even the upper ends stay below a third.

0 · 4 · 3 — abstentions of 31 when the observing instruments were removed (Gemini 3.8 Live, Opus 5, GPT-6 Astra)

36 / 37 — fault-free runs Astra called anomalous, with a physically coherent argument: it did not know this plant's normal

8–97 % — false-alarm range across models: behaviour transferred to real data, rates did not

Truth from the plant's expert annotations. The first run had a wrong instrument legend; the registered rerun with the corrected legend left abstention unchanged and did not remove the false alarms, so the alarms are the models' judgement, not a harness artefact.

HOW IT WAS TESTED

- **21 pre-registrations**: predictions, reading rules and stop conditions fixed before the first scored call; every deviation logged; failed predictions reported beside the ones that held.
- Differences under 3 items are not interpreted; key claims rest on three runs (pass^3); a same-condition control arm for every new mechanism.
- Seven models from four vendors (Claude Opus 5 / 5.5, Gemini 3.8 Live / Flash / 3.1 Pro, GPT-6 Astra, GLM 5.3 Flash), one runner, every request and response kept. Across models only the direction is read.
- Before blaming a model: check the measurement, then the task statement. Two harness bugs and one unstated scoring rule each first looked like model behaviour.

Data

DATA	ITEMS	TRUTH
LiveLab Core: simulated CVD tube furnace, faults injected from a fault library	80 (30 twin pairs, 10 fault-free, 10 all sensors)	rule-computed
Real batch distillation plant, public 119-run dataset (Zenodo 17395543, CC BY)	99 (31 full, 31 evidence removed, 37 fault-free)	plant experts' annotations
Self-driving-lab inspection photographs (published, CC BY)	100	published labels

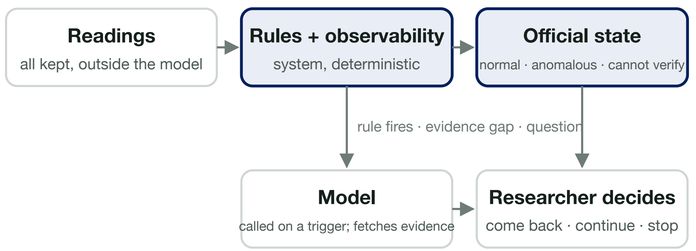
Code, registrations, logs github.com/dekunsun/livelab

Preprint and results dekunsun.github.io/livelab

What this does not show, and what it argues for

- **No real-lab rates.** Core is simulated; on real plant data the behaviour transferred but the rates did not. Stating the rules is one run per model: a candidate, not a qualification. No configuration has passed the registered gate (private hold-out, three runs).
- **Design response.** The system, not the model, computes what the installed instruments can settle and owns the official state; NORMAL means verified. Deterministic rules keep running on what remains; the model explains, triages and fetches evidence on demand, and can never turn "cannot verify" into "normal". Its own alarms are shown as "model flag, pending review". Evidence is passed raw, never as a summary: every text hand-off lost accuracy (inspection photographs: Gemini 74 → 51, Opus 5 57 → 23 of 100).
- **Next steps.** A partner lab's logs and interviews (do researchers see false reassurance as a failure worth a guardrail? untested); the hold-out three times; a second real dataset for 2.0, since the current one is exhausted; shadow mode before any alert reaches a person.

Monitoring workflow



The model never rewrites the state; its own alarms are "pending review".

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AI FOR SCIENCE · EVIDENCE AND ITS BOUNDARIES

A field guide to AI for science: research directions, companies, evaluation specs and interviews, kept honest with evidence ladders.

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